kinematics

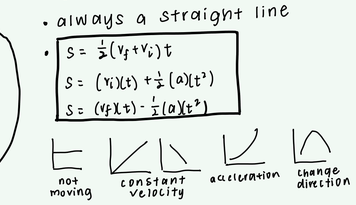
SCALAR QTY

VECTOR QTY

distance (m)

position (m)

- always a straight line
- $s = \frac{1}{2}(v_f + v_i)t$
  $s = (v_i)(t) + \frac{1}{2}(a)(t^2)$
  $s = (v_f)(t) - \frac{1}{2}(a)(t^2)$



Motion maps

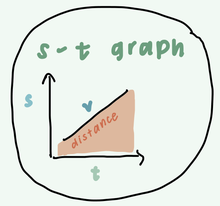


, represents movements at different times

Speed (m/s)

average speed	instantaneous speed
$\frac{\text{total distance}}{\text{total time}}$	rate of change of distance at each instant

s - t graph



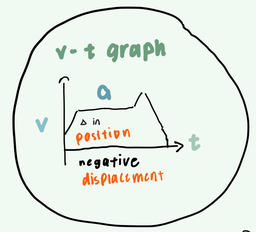
area

gradient

time (s)

Instantaneous
- at a certain point in time

v - t graph



Velocity (m/s)

$\frac{\text{disp}}{t}$

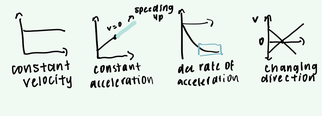
- speed with direction
- can be + or −

Instantaneous velocity

rate of change of displacement at each instant

$v_f^2 = v_i^2 + 2(a)(s)$

gradient



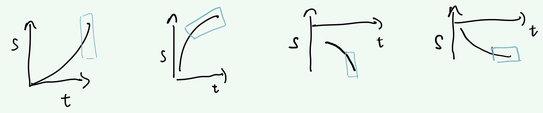
OBJECT MOVING / CONSTANT ACCELERATION

+ direction		- direction	
speed ↑ (acceleration)	slow ↓ (deacceleration)	speed ↑ (acceleration)	slow ↓ (deacceleration)

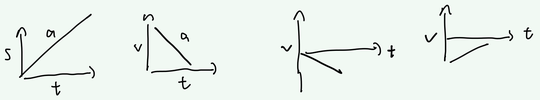
x - t

① speed (slope)
② direction

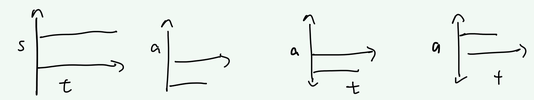




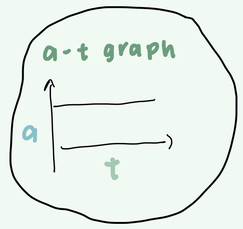
v - t



a - t



a - t graph



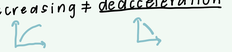
Acceleration (m/s²)

- rate of change in velocity
- be speeding up or slowing down

$a = \frac{v_f - v_i}{t}$

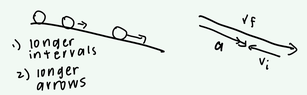
v ↑	+
v ↓	-

decreasing ≠ slowing down deacceleration



acceleration due to gravity: $-9.81\ ms^{-2}$

Speeding up



Slowing down

